```
String str1 = sc.nextLine();
 System.out.print("Enter second string: ");
 String str2 = sc.nextLine();
 Con con = new Con();
Comp comp = new Comp();
System.out.println("\nString Operations:");
System.out.println("-------------------------");
 con.concatenate(str1, str2);
 comp.compare(str1, str2);
 sc.close();
 }
 }
```

Slip 5

1) Write a java program to display the contents of a file in reverse order.

Step 1: create txt file= vim test.txt

Step 2: Main program vim ReverseFile.java

```
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;

public class ReverseFile {
   public static void main(String args[]) {

      Scanner sc = new Scanner(System.in);

      System.out.print("Enter file name: ");
      String fileName = sc.nextLine();
```

```
        String content = "";

        try {
            FileReader fr = new FileReader(fileName);

            int ch;

            while ((ch = fr.read()) != -1) {
                content = content + (char) ch;
            }

            fr.close();

            System.out.println("\nFile contents in reverse order:");
            System.out.println("--------------------------------");

            for (int i = content.length() - 1; i >= 0; i--) {
                System.out.print(content.charAt(i));
            }

        } catch (IOException e) {
            System.out.println("Error: File not found or cannot be read.");
        }

        sc.close();
    }
}
```

2) Write a java program to create base class Employee (empcode,empname). Derive the classes anager(designation,club_dues), Scientist (deptname,publications) from Employee class accept the details of n employees and to display the information.

```
import java.util.Scanner;


class Employee {
   int empcode;
   String empname;


   Employee(int empcode, String empname) {
      this.empcode = empcode;
      this.empname = empname;
   }


   void display() {
      System.out.println("Employee Code : " + empcode);
      System.out.println("Employee Name : " + empname);
   }
}


class Manager extends Employee {
   String designation;
   double club_dues;


   Manager(int empcode, String empname, String designation, double club_dues) {
      super(empcode, empname);
      this.designation = designation;
      this.club_dues = club_dues;
```

```
    }

    void display() {
        super.display();
        System.out.println("Designation   : " + designation);
        System.out.println("Club Dues     : " + club_dues);
        System.out.println("--------------------------------");
    }
}

class Scientist extends Employee {
    String deptname;
    String publications;

    Scientist(int empcode, String empname, String deptname, String publications) {
        super(empcode, empname);
        this.deptname = deptname;
        this.publications = publications;
    }

    void display() {
        super.display();
        System.out.println("Department    : " + deptname);
        System.out.println("Publications  : " + publications);
        System.out.println("--------------------------------");
    }
}
public class EmployeeDemo {
```

```
public static void main(String args[]) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number of employees: ");
    int n = sc.nextInt();
    sc.nextLine();
    Employee employees[] = new Employee[n];
    for (int i = 0; i < n; i++) {
        System.out.println("\nEmployee " + (i + 1));
        System.out.println("1. Manager");
        System.out.println("2. Scientist");
        System.out.print("Enter choice: ");
        int choice = sc.nextInt();
        sc.nextLine()
        System.out.print("Enter Employee Code: ");
        int empcode = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Employee Name: ");
        String empname = sc.nextLine();
        if (choice == 1) {
            System.out.print("Enter Designation: ");
            String designation = sc.nextLine();
            System.out.print("Enter Club Dues: ");
            double club_dues = sc.nextDouble();
            sc.nextLine();
            employees[i] = new Manager(
                empcode, empname, designation, club_dues
            );
        } else if (choice == 2) {
```

```
            System.out.print("Enter Department Name: ");
            String deptname = sc.nextLine();
            System.out.print("Enter Publications: ");
            String publications = sc.nextLine();
            employees[i] = new Scientist(
               empcode, empname, deptname, publications
            );
          } else {
            System.out.println("Invalid choice.");
            i--;
          }
        }

        System.out.println("\nEmployee Information");
        System.out.println("==============================");
        for (int i = 0; i < n; i++) {
          employees[i].display();
        }
        sc.close();
    }
}
```

OR

2) Write a java program to create an abstract class Shape with methods area & volume. Derive two classes Sphere (radius), Cylinder (radius, height) from it. Calculate area and volume of both. (Use Method Overriding).

```
import java.util.Scanner;

abstract class Shape {
   abstract void area();
```

```
    abstract void volume();
}

class Sphere extends Shape {
    double radius;

    Sphere(double radius) {
        this.radius = radius;
    }

    void area() {
        double a = 4 * Math.PI * radius * radius;
        System.out.println("Area of Sphere = " + a);
    }

    void volume() {
        double v = (4.0 / 3.0) * Math.PI * radius * radius * radius;
        System.out.println("Volume of Sphere = " + v);
    }
}

class Cylinder extends Shape {
    double radius;
    double height;

    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
```

```
    }

    void area() {
        double a = 2 * Math.PI * radius * (radius + height);
        System.out.println("Area of Cylinder = " + a);
    }

    void volume() {
        double v = Math.PI * radius * radius * height;
        System.out.println("Volume of Cylinder = " + v);
    }
}

public class ShapeDemo {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter radius of Sphere: ");
        double sphereRadius = sc.nextDouble();

        System.out.print("Enter radius of Cylinder: ");
        double cylinderRadius = sc.nextDouble();

        System.out.print("Enter height of Cylinder: ");
        double cylinderHeight = sc.nextDouble();

        Sphere sphere = new Sphere(sphereRadius);
```